



U.S. General Services
Administration

Chapter 5.7: Episodic Memory

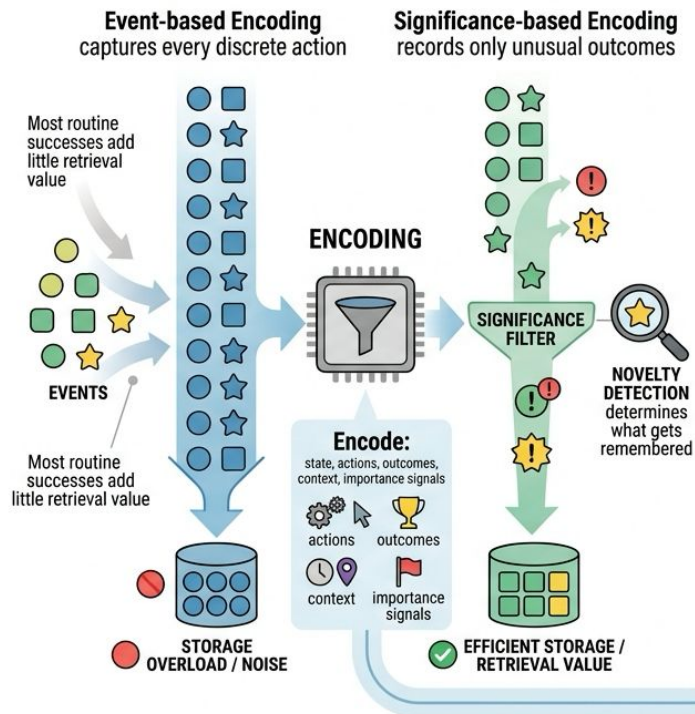
US General Service Administration
AI Community of Practice - March 2026

Three Memory Types Agents Confuse

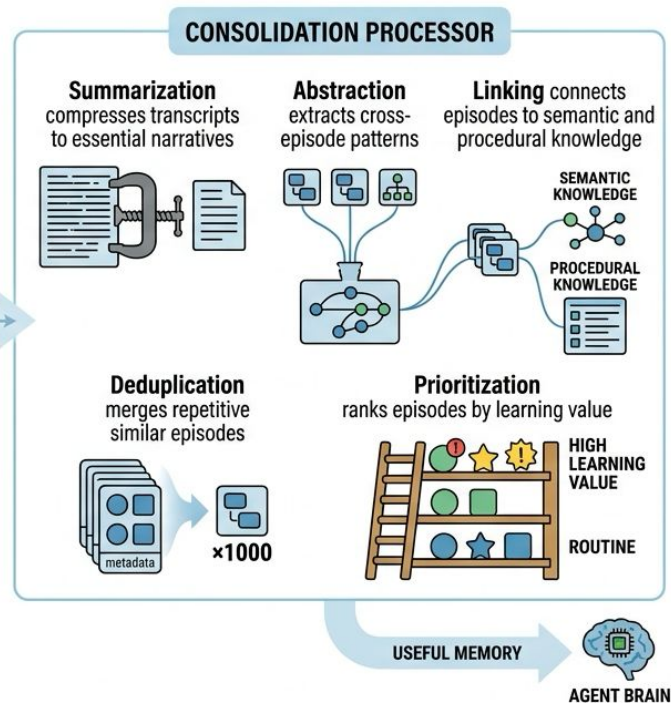
- Semantic memory holds timeless general knowledge
 - Working memory is the volatile current-session context
 - Procedural memory encodes "how to" workflows
 - Episodic memory records specific events with timestamps
 - Mixing them up causes architectural disasters
- 

Encoding and Consolidation

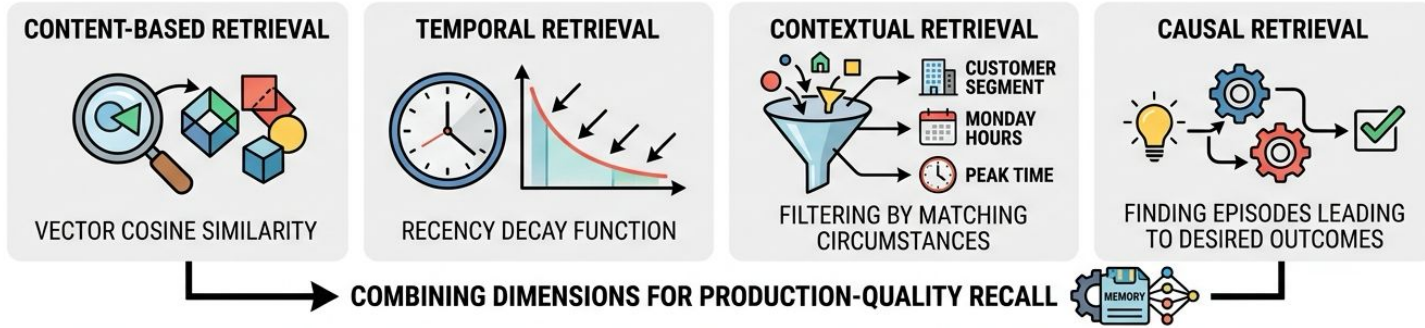
ENCODING -- DECIDING WHAT DESERVES STORAGE



CONSOLIDATION -- FROM RAW EVENTS TO USEFUL MEMORY



Retrieval Mechanisms

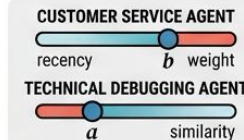
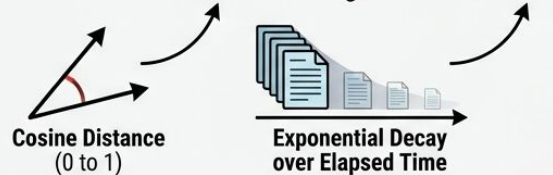


THE SCORING FORMULA: RANKING RETRIEVED MEMORIES

TUNING WEIGHTS (a, b, c) IS DOMAIN-SPECIFIC

$$\text{retrieval_score} = a \times \text{similarity} + b \times \text{recency} + c \times \text{importance}$$

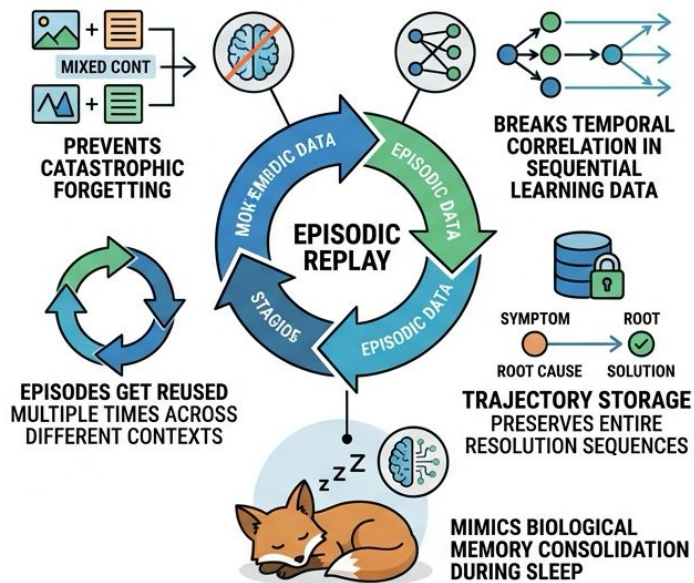
Icons for **Feedback** (thumbs up), **Novelty** (star), and **Outcome Quality** (stack of coins) are shown next to the formula.



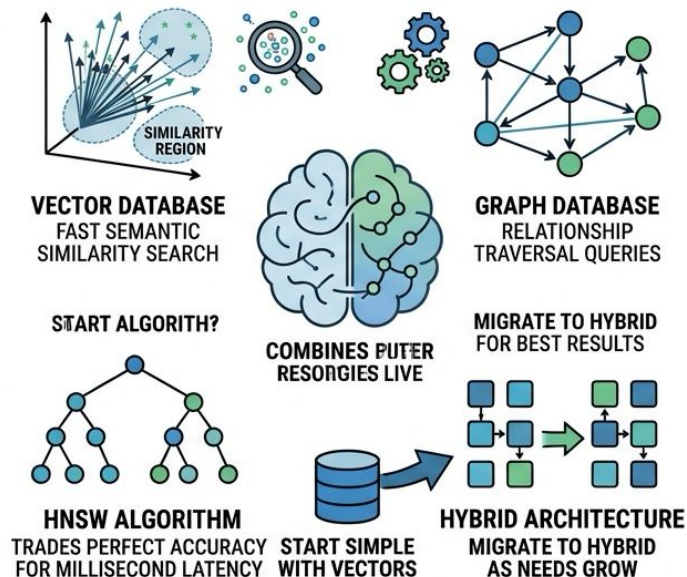
RETRIEVAL IS A RANKING PROBLEM, NOT JUST SIMILARITY SEARCH

Experience Replay and Storage

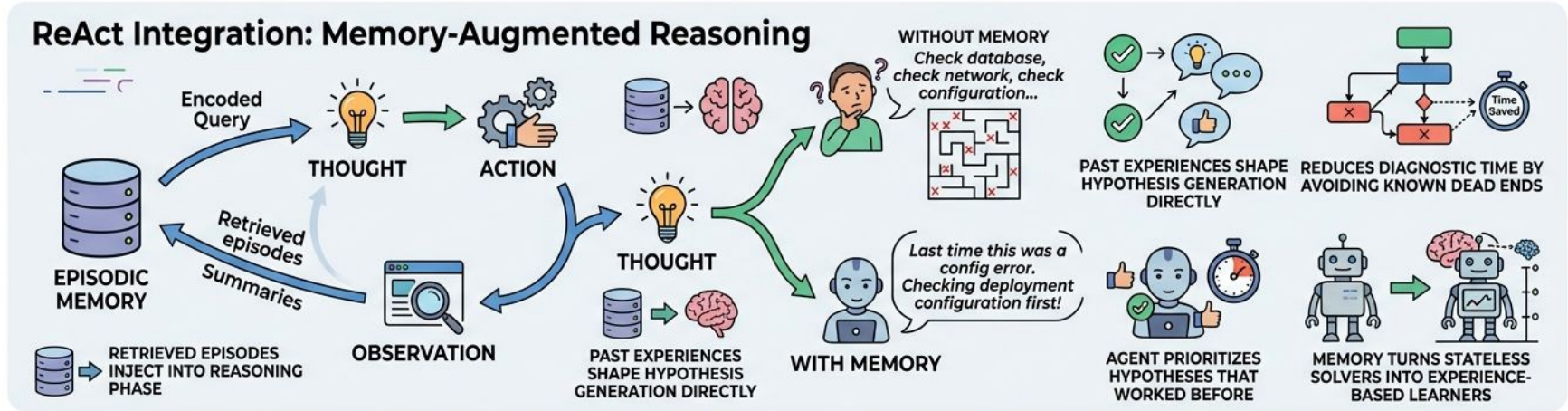
LEARNING OFFLINE FROM PAST EPISODES



WHERE MEMORIES LIVE



ReAct Integration -- Memory-Augmented Reasoning



- Retrieved episodes inject into the reasoning phase of ReAct
- Past experiences shape hypothesis generation directly
- Agent prioritizes hypotheses that worked before
- Reduces diagnostic time by avoiding known dead ends
- Memory turns stateless solvers into experience-based learners

Production Applications

PERSONALIZED ASSISTANTS



PERSONALIZED ASSISTANTS

Maintain context across sessions
(e.g., Denver trip details)

RL ROBOTS



RL ROBOTS

Use replay to learn new skills
(boxes) without forgetting
old ones (small objects)

DEBUGGING AGENTS



DEBUGGING AGENTS

Learn diagnostic shortcuts
from past trajectories
(e.g., faster timeout fixes)

DELIVERY ROBOT FLEETS



DELIVERY ROBOT FLEETS

Share obstacle episodes
collectively (e.g., rerouting
all for construction)

FRAUD DETECTION



FRAUD DETECTION

Build personalized baselines
per customer to catch
anomalies

- Personalized assistants maintain context across sessions
- RL robots use replay to learn new skills without forgetting old ones
- Debugging agents learn diagnostic shortcuts from past trajectories
- Delivery robot fleets share obstacle episodes collectively
- Fraud detection builds personalized baselines per customer

Production Pitfalls -- What Breaks at Scale

- Embedding models have domain-specific blind spots
 - Context window overflow from retrieving too many episodes
 - Consolidation is lossy compression -- unique details vanish
 - Contradictory episodes require explicit conflict resolution
 - Storing episodes is memorization, not generalization
- 

Key Takeaways and Bridge to Semantic Memory

- Episodic memory makes agents learn from lived experience
 - Encode selectively, consolidate aggressively, retrieve smartly
 - Combine similarity, recency, and importance in retrieval scoring
 - Experience replay prevents forgetting and enables pattern discovery
 - Next: Chapter 5.8 explores semantic memory for general knowledge
- 